📄 Data Science Project Kickoff Checklist

A senior-practitioner tool for decision-first, low-rework project starts
Purpose: Ensure early decisions align the entire pipeline before any irreversible choices are made.
You do **not** need all answers perfectly. You **do** need them explicitly acknowledged.

1️⃣ Decision Being Supported (Required)

Decision statement:
*We are deciding **[action]** under **[constraints]** with uncertainty about **[unknowns]**.*
Who makes this decision?

- ☐ Human
- ☐ System
- ☐ Hybrid Role / team: _____

Decision frequency:

- ☐ One-time
- ☐ Periodic
- ☐ Continuous

🔴 **Red flag:** "We're not sure yet — we just want insights."

2️⃣ Consequences & Irreversibility (Risk Surface)

If the decision is wrong, what happens?
☐ Financial cost
☐ Trust / credibility loss
☐ Opportunity cost
☐ Compliance / reputational risk
☐ Other: _____
Which mistake is worse?
☐ Acting when we shouldn't (false positives)
☐ Not acting when we should (false negatives)
What cannot be undone once action is taken?
☐ Money spent
☐ Customers contacted

☐ Policies triggered
☐ Other: _____
🔴 **Red flag:** "We'll know it when we see it."

3️⃣ Constraints (Design Inputs)

Hard constraints (non-negotiable):
☐ Budget
☐ Time
☐ People / operations
☐ Policy / legal
☐ Other: _____
Actions explicitly out of scope (even if data suggests them):
test

Decisions that must remain human:
test

🔴 **Red flag:** "Let's see what the data says first."

4️⃣ Evidence & Trust Calibration (Ingestion Lens)

Evidence Source	Type (Authoritative / Proxy / Heuristic)	Owner	Confidence (High / Med / Low)

Evidence currently trusted most (even if imperfect):
test

Evidence known to be misleading or fragile:
test

🔴 **Red flag:** "The data should be fine — it's in the system."

5️⃣ Uncertainty & Assumptions (Make Them Visible)

☐ **What do we *not* know that materially affects the decision?**

- Timing
- Causality
- Coverage gaps
- Biases

☐ **What assumptions are currently being made (explicitly or implicitly)?**

- About behavior
- About labels
- About stability over time

☐ **Which assumptions would be dangerous if wrong?**

test

🔴 **Red flag:** "We'll clean that up later."

6️⃣ Success Criteria (Without Metrics Yet)

What would make you confident enough to use this?

☐ Clear explanations
☐ Comparisons / baselines
☐ Guardrails
☐ Known failure modes
☐ Other: _____

What would make you *not* trust it?

test

What does 'good enough' look like for a first version?

test

🔴 **Red flag:** "High accuracy."

7️⃣ Explicit Non-Goals (This Prevents Scope Creep)

This system will not:

☐ Make automated decisions
☐ Optimize for all use cases
☐ Replace human judgment
☐ Answer unrelated questions

Explicitly out of scope:
test

🔴 **Red flag:** "Let's keep it flexible."

8️⃣ End-of-Kickoff Sanity Check (Always Do This)

Before leaving the conversation, confirm:
☐ Decision can be stated in one sentence
☐ Risks and constraints are acknowledged
☐ Uncertainty is acceptable and expected
☐ This is decision support, not an oracle

If any box is unchecked → do not move downstream yet.

🧠 How to Use This in Practice (Low-Friction)

- You do **not** need to ask every question verbatim
- Many answers come from listening, not interrogating
- You can fill this out **after** the meeting based on notes
- Treat it as a *thinking tool*, not a form to hand over

Even completing **60–70%** of this checklist dramatically reduces redesign risk.

Why This Works

This checklist:

- locks alignment *before* irreversible pipeline choices
- forces risk to surface early
- naturally guides cleaning, features, modeling, and evaluation
- gives you cover later when tradeoffs are questioned

Most DS project failures happen **before code exists**. This is how you prevent that — quietly and professionally.